ISEL \ DEETC

MEIM | MEIC

AMD (MLDM – Machine Learning and Data Mining) – Module of Practice #03

Paulo Trigo Silva

---

1. Install and test "Orange Data Mining Framework" and "Python"

The steps to download/install "Orange Data Mining Framework" are as follows.

a) If you have a **64bit** architecture download directly from the Orange site:

`http://orange.biolab.si/download/`
Windows, choose: "`Orange3-3.39.0-x86_64.zip`".
MacOS[X], according to your architecture choose [X]: "`Orange3-3.39.0-Python3*-[X].dmg`"

for more details see "`https://github.com/biolab/orange3/blob/master/README.md`"

b) **ONLY** If you have a **32bit** architecture (*this was my case!*):

1. download a 32bit older version from:

   `https://download.biolab.si/download/files/`
   "`Orange3-3.21.0-Python36-win32.exe`" seems to be the last 32bit version.

2. this Orange version version (3.21.0) is bundled with Python3.6.4

   this Orange version installs the framework within a "python virtual environment"; for more information on "virtual environment": `https://virtualenv.pypa.io/en/stable/`.
   So, you will find the installation in "(…)\Orange" folder.
   note: there are useful example datasets in "(...)\Orange\Lib\site-packages\Orange\datasets"

3. test if your installation can properly use the Orange libraries:

   open a new "Command Prompt" window
   change current folder to the installation; cd c:\(…)\Orange
   initiate the Orange virtual environment; execute "Orange Command Prompt" file
   launch Python interpreter (i.e., write python):

   ```
   >>> import Orange
   >>>
   ```

   To quit the Python environment write: `quit()`

   In case you still problems with the Orange installation visit:
   `https://datascience.stackexchange.com/tags/orange/info/`

4. follow the instructions in "**`_USAGE_virtual-environment-for-python.txt`**" file regarding the usage of "Python Virtual Environment

If you are using "Visual Studio Code" or "PyCharm" as your IDE (integrated development environment) and you are having problems with the "`import Orange`" statement (i.e., using Orange library), then configure the IDE to use the "`python.exe`" that is bundled in Orange. So, check your Orange installation folder (usually `c:\Program Files\Orange`) and set the IDE to use the "`python.exe`" that is available within that folder.

AMD (MLDM – Machine Learning and Data Mining) – Module of Practice #03

Paulo Trigo Silva

2. Read the dataset using "Orange Canvas" – the graphical mode

a) Copy the file, "dataset_3RowHeader.txt", built in the previous practical class to this class's folder.

b) Execute the "Orange Canvas" application: "\Start\Orange3\Orange Canvas".

c) Select the "Data" separator. Drag the "File" icon into the white part of the canvas.

d) Make a "double-click" in the "File" icon and choose the "Data File" to point to the folder that contains the "dataset_3RowHeader.txt" file. Notice that the file is not available for selection! **To fix the problem** change the file extension **".txt" to ".tab"** and select the file.

e) Take a look at the file "dataset_finalFormat.tab" provided with this practical lesson. If necessary adapt the script developed in previous practical lesson in order to generated a file with this format.

f) Drag the "Data Table" icon into the white part of the canvas; "double-click" on the icon and notice the "info \ no data on input" (on the top-left side of the window).

g) Make a connection between "File" and "Data Table" icons. To connect two processes, represented by icon A and icon B, select a connector on the right side (output pipe) of A and drag it into a left connector of B (input pipe). A connection is a data flow from an output pipe into an input pipe.

h) See the "Data Table" output. Connect the "Data Table" process to a "Save Data" process.

i) Point to the connector line and "double-click" (or "right-click" and choose "Reset Signals"). Notice that "Data Table" process has two output available ports; explore deleting established connection and activating the other one.

j) Configure the "Save" process so that it generates a ".csv" ("comma separated values") after some data transformations (e.g., via a pipeline of the processes explored in the previous items).

k) Use Excel to read the file that was generated in the previous item.

l) Explore the "Data \ Feature Statistics" process.

m) Explore the "Transform \ Select Columns" process.

n) Explore the "Transform \ Select Rows" process.

o) Explore the "Transform \ Purge Domain" process.

p) Explore the "Transform \ Discretize" process.

q) Save your work (\File \ Save As…) into an "Orange Schema" so that it can be used again later.

3. Reuse an "Orange Schema"

a) Open the "`z01_schema.ows`"; double-click on the file or open it within the "Orange Canvas".

b) Explore the schema and make sure that the "*Save (z_out_xx)*" widgets generate ".tab" files with the same information as in the provided "`z_out_xx.txt`" files. *Hint*: explore the options available in each widget and each link (right-click over the link and "Reset Signals").

AMD (MLDM – Machine Learning and Data Mining) – Module of Practice #03

Paulo Trigo Silva

4. Start working with "`lenses`" dataset in Orange – graphic mode

Consider the dataset: "`lenses.tab`" (in folder "`_dataset`").

a) Execute the "Orange Canvas", choose the "File" icon and read the dataset.

Note: we may consider an "Orange" icon as an "operator" of a visual programming language; so, instead of calling "icon" we may call "operator".

b) Connect the "File" operator with "Data Table" and visualize all the information about the dataset.

5. [before-proceeding] – a Python "crash-course"

Consider the file: "`a00_python_crash_course.py`".

a) Open the "`a00_python_crash_course.py`" and comment everything. *Hint*: if you are using IDLE, in order to comment a block of lines, select the lines and then select the menu option "\Format\Comment Out Region".

b) Uncomment each section (e.g., Strings, Numbers, Boolean, Multiple Assignment, No value, etc) at a time and explore the concept e that section. We can search for additional information for example in the "p01_aByteOfPython_v3.pdf" document.

c) In the end make sure that the value computed in the "main" function gets printed!

d) Now, encapsulate each section (e.g., Strings, Numbers, Boolean, etc) in a function and make sure that each of those new functions are called from the "main" function.

6. Read a dataset with Orange – programmatic mode

Consider the file: "`a01_datasetRead.py`".

e) Open the "`a01_datasetRead.py`" and comment everything except for the first lines of code where a dataset is read. Execute the code. *Hint*: if you are using IDLE, in order to comment a block of lines, select the lines and then select the menu option "\Format\Comment Out Region".

f) Add to the code a line with `print( dataset )` and execute.

g) Now add a line to print `dataset.domain` and execute.

h) Now add a line to print `dataset.domain.variables` and execute.

i) Now add a line to print `dataset.domain.attributes` and execute.

j) Now add a line to print `dataset.domain.class_var` and execute.

k) Now add a line to print the `dir( dataset.domain.class_var )` and execute.

l) Eliminate the comments on the code that analyses the `dataset.domain.attributes`. Execute and explain the difference from the `dataset.domain.variables` structure.

m) Eliminate the remaining comments, execute and explain the result.

AMD (MLDM – Machine Learning and Data Mining) – Module of Practice #03

Paulo Trigo Silva

7. The (second) "Kick-Off" of Final Project A

a) Consider "final project A" (cf., moodle, "Final Project" folder). Develop "Project Item: 5".

b) Use the Orange Canvas to create a process that loads and (eventually) applies transformations to the dataset exported from PostgreSQL database. Save your work as an "Orange schema".

c) Create a Python automatism to generate the dataset. *Suggestion*: extend the "`_goPy.py`" script provided in the previous practical class (cf. folder "`scripts`").

d) Create a Python application to extract the metadata (features and class attributes along with the corresponding domains and values in the dataset) that is necessary to implement the 1R method.